PC1> ip 192.168.1.1/24

Checking for duplicate address...

PC1 : 192.168.1.1 255.255.255.0

PC2> ip 192.168.1.2/24

Checking for duplicate address...

PC2 : 192.168.1.2 255.255.255.0

ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=0.291 ms

84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=0.343 ms

84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=0.313 ms

84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=0.261 ms

84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=0.267 ms

PC1>

ARP-запрос:

Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

  Destination: Broadcast (ff:ff:ff:ff:ff:ff)

  Source: Private_66:68:00 (00:50:79:66:68:00)

  Type: ARP (0x0806)

  [Stream index: 0]

  Padding: 000000000000000000000000000000000000

  Frame check sequence: 0x00000000 [unverified]

  [FCS Status: Unverified]

Address Resolution Protocol (request)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)

Sender IP address: 192.168.1.1

Target MAC address: 00:00:00:00:00:00

Target IP address: 192.168.1.2

ARP-ответ:

Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00)

Destination: Private_66:68:00 (00:50:79:66:68:00)

.... ..0. .... = LG bit: Globally unique address (factory default)

....0 .... = IG bit: Individual address (unicast)

Source: Private_66:68:01 (00:50:79:66:68:01)

.... ..0. .... = LG bit: Globally unique address (factory default)

....0 .... = IG bit: Individual address (unicast)

Type: ARP (0x0806)

[Stream index: 2]

Padding: 000000000000000000000000000000000000

Frame check sequence: 0x00000000 [unverified]

[FCS Status: Unverified]

Address Resolution Protocol (reply)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: reply (2)

Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)

Sender IP address: 192.168.1.2

Target MAC address: Private_66:68:00 (00:50:79:66:68:00)

Target IP address: 192.168.1.1

Захват производился на всех линках топологии: линк №1 (ПК1–коммутатор) и линк №2 (коммутатор–ПК2). Структура ARP-пакетов на обоих линках идентична!

```
R1#enable

R1#configure terminal

Enter configuration commands, one per line.  End with CNTL/Z.

R1(config)#interface FastEthernet0/0

R1(config-if)# ip address 192.168.1.1 255.255.255.0

R1(config-if)# no shutdown

R1(config-if)# exit

R1(config)#interface FastEthernet0/1

                    ^

% Invalid input detected at '^' marker.


R1(config)# ip address 192.168.2.1 255.255.255.0

                     ^

% Invalid input detected at '^' marker.


R1(config)# no shutdown

% Incomplete command.


R1(config)# end
```

```
R1#
*Mar  1 00:02:40.087: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0            192.168.1.1     YES manual up                    up
FastEthernet1/0            unassigned      YES unset  administratively down down
Ethernet2/0                unassigned      YES unset  administratively down down
Ethernet2/1                unassigned      YES unset  administratively down down
Ethernet2/2                unassigned      YES unset  administratively down down
Ethernet2/3                unassigned      YES unset  administratively down down
Serial3/0                  unassigned      YES unset  administratively down down
Serial3/1                  unassigned      YES unset  administratively down down
Serial3/2                  unassigned      YES unset  administratively down down
Serial3/3                  unassigned      YES unset  administratively down down
R1#
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface FastEthernet1/0
R1(config-if)# ip address 192.168.2.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# end
*Mar  1 00:03:32.387: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar  1 00:03:33.387: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R1(config-if)# end
R1#
*Mar  1 00:03:36.031: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip interface brief
```

```
Interface              IP-Address      OK? Method Status            Protocol
FastEthernet0/0        192.168.1.1     YES manual up                up
FastEthernet1/0        192.168.2.1     YES manual up                up
Ethernet2/0            unassigned      YES unset  administratively down down
Ethernet2/1            unassigned      YES unset  administratively down down
Ethernet2/2            unassigned      YES unset  administratively down down
Ethernet2/3            unassigned      YES unset  administratively down down
Serial3/0              unassigned      YES unset  administratively down down
Serial3/1              unassigned      YES unset  administratively down down
Serial3/2              unassigned      YES unset  administratively down down
Serial3/3              unassigned      YES unset  administratively down down
R1#
R1#show ip interface brief
Interface              IP-Address      OK? Method Status            Protocol
FastEthernet0/0        192.168.1.1     YES manual up                up
FastEthernet1/0        192.168.2.1     YES manual up                up
Ethernet2/0            unassigned      YES unset  administratively down down
Ethernet2/1            unassigned      YES unset  administratively down down
Ethernet2/2            unassigned      YES unset  administratively down down
Ethernet2/3            unassigned      YES unset  administratively down down
Serial3/0              unassigned      YES unset  administratively down down
Serial3/1              unassigned      YES unset  administratively down down
Serial3/2              unassigned      YES unset  administratively down down
Serial3/3              unassigned      YES unset  administratively down down
R1#

PC1> ping 192.168.2.2

192.168.2.2 icmp_seq=1 timeout
```

84 bytes from 192.168.2.2 icmp_seq=2 ttl=63 time=11.751 ms

84 bytes from 192.168.2.2 icmp_seq=3 ttl=63 time=15.793 ms

84 bytes from 192.168.2.2 icmp_seq=4 ttl=63 time=15.581 ms

84 bytes from 192.168.2.2 icmp_seq=5 ttl=63 time=16.013 ms

PC1>

ARP-запрос:

Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Destination: Broadcast (ff:ff:ff:ff:ff:ff)

Source: Private_66:68:00 (00:50:79:66:68:00)

Type: ARP (0x0806)

[Stream index: 2]

Padding: 000000000000000000000000000000000000

Frame check sequence: 0x00000000 [unverified]

[FCS Status: Unverified]

Address Resolution Protocol (request)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)

Sender IP address: 192.168.1.2

Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)

Target IP address: 192.168.1.1

ARP-ответ:

Ethernet II, Src: cc:01:37:d6:00:00 (cc:01:37:d6:00:00), Dst: Private_66:68:00 (00:50:79:66:68:00)

Destination: Private_66:68:00 (00:50:79:66:68:00)

.... ..0. .... = LG bit: Globally unique address (factory default)

....0 .... = IG bit: Individual address (unicast)

Source: cc:01:37:d6:00:00 (cc:01:37:d6:00:00)

.... ..0. .... = LG bit: Globally unique address (factory default)

....0 .... = IG bit: Individual address (unicast)

Type: ARP (0x0806)

[Stream index: 3]

Padding: 000000000000000000000000000000000000

Address Resolution Protocol (reply)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: reply (2)

Sender MAC address: cc:01:37:d6:00:00 (cc:01:37:d6:00:00)

Sender IP address: 192.168.1.1

Target MAC address: Private_66:68:00 (00:50:79:66:68:00)

Target IP address: 192.168.1.2

ICMP Echo Request:

Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: cc:01:37:d6:00:00 (cc:01:37:d6:00:00)

Type: IPv4 (0x0800)

Internet Protocol Version 4, Src: 192.168.1.2, Dst: 192.168.2.2

0100 .... = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 84

Identification: 0xdd01 (56577)

000. .... = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 64

Protocol: ICMP (1)

Header Checksum: 0x1953 [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.1.2

Destination Address: 192.168.2.2

[Stream index: 0]

Internet Control Message Protocol

Type: Echo (ping) request (8)

Code: 0

Checksum: 0x1e2e [correct]

[Checksum Status: Good]

Identifier (BE): 477 (0x01dd)

Identifier (LE): 56577 (0xdd01)

Sequence Number (BE): 1 (0x0001)

Sequence Number (LE): 256 (0x0100)

ICMP Echo Reply:

Ethernet II, Src: cc:01:37:d6:00:00 (cc:01:37:d6:00:00), Dst: Private_66:68:00 (00:50:79:66:68:00)

Internet Protocol Version 4, Src: 192.168.2.2, Dst: 192.168.1.2

Time to Live: 63

Internet Control Message Protocol

Type: Echo (ping) reply (0)

Code: 0

Захват трафика производился на всех линках топологии в соответствии с требованием nb!. На линке №1 ARP-обмен происходит между ПК1 и роутером. ПК1 запрашивает MAC-адрес шлюза

(192.168.1.1) и получает ответ. На линке №2 роутер запрашивает MAC-адрес ПК2 (192.168.2.2) и также получает ответ. MAC-адреса на разных линках отличаются. Структура запросов и ответов на линке 2 имеет аналогичную структуру.